Problem Set 3: Econ 5030

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1 Basic Cournot duopoly

Let begin by considering a Cournot duopoly. Each firm i has the cost function $C(q_i) = cq_i^2/2$, where $c > 0$. Market inverse $P(Q) = 200 - Q$.

1.a Write the profit of the arbitrary firm i.

1.b Find the best response function for firm i and graph it on the $q_i \times q_j$ plane.

1.c Find the Cournot equilibrium quantity q^c and market price $p^c = P(2q^c)$

1.d Graph the market outcome on the $p \times Q$ plane.

1.e Find an expression for the each firm's profits, the Consumer Surplus (CS),Total Surplus (TS) and Dead weight Loss (DWL).

2 Basic Cournot oligopoly

Let now consider a Cournot oligopoly with n firms. Each firm i has the cost function $C(q_i) = cq_i^2/2$, where $c > 0$. Market inverse $P(Q) = 200 - Q$.

2.a Write the profit of the arbitrary firm i.

2.b Find the best response function for firm i and graph it on the $q_i \times Q_{-i}$ plane.

2.c Find the Cournot equilibrium quantity q^c and market price $p^c = P(nq^c)$

2.d Graph the market outcome on the $p \times Q$ plane.

2.e Find an expression for the each firm's profits, the Consumer Surplus (CS), Total Surplus (TS) and Dead weight Loss (DWL).

3 Cournot with Product Differentiation

Take the inverse demand system:

$$\begin{aligned} P_1(q_1, q_2) &= 18 - q_1 - \gamma q_2, \\ P_2(q_2, q_1) &= 18 - q_2 - \gamma q_1. \end{aligned}$$

Assume both firms have NO COST of production.

Now let us look at the Cournot solution for general γ between zero an one.

2.a Write out each firm's profit function and label them $\Pi_1(\gamma)$ and $\Pi_2(\gamma)$.

2.b Find each firm's best response function by solving $\frac{\partial \Pi_1(\gamma)}{\partial q_1} = 0$ for $q_1^\gamma(q_2)$ and $\frac{\partial \Pi_2(\gamma_2, \gamma_1)}{\partial q_2} = 0$ for $q_2^\gamma(q_1)$.

2.c Graph the best response functions of the firms' on $q_1 \times q_2$ space.

2.d Find the equilibrium quantities and prices and label them $q_1(\gamma)$, $p_1(\gamma)$ and $q_2(\gamma)$, $p_2(\gamma)$.

4 Price competition

Suppose that thre are 2 firms in a Bertrand oligopoly with homogenous products. Each firm i has th cost of production $c(q_i) = q_i^2/2$. The industry demand function is the rectangle

$$D(p) = \begin{cases} 0 & \text{if } p > 10 \\ 10 & \text{if } p \leq 10 \end{cases}$$

The if firms price the same then demand is split evenly. Each firm i has th cost of production $c(q_i) = q_i^2/2$. Based on this cost structure a firm will never supply more than their supply function $s(p_i) = p_i$. using this fact we can construct residual demand and firm profit. Firm profit is the piecewise function for prices in the interval

$[0, 10]$ (I have done all the work for you here):

$$\pi_i(p_i, p_j) = \begin{cases} p_i \min\{p_i, 10 - p_j\} - (\min\{p_i, 10 - p_j\})^2 / 2 & p_i > p_j \\ p_i 5 - 12.5 & p_i = p_j > 5 \\ p_i^2/2 & p_i = p_j \leq 5 \\ p_i^2/2 & p_i < p_j \end{cases}$$

1. Because for most prices p_j the best response is not well-defined, instead find the argument of the supremum of the profit responding to the other firm's price. Formally find
$$\phi(p_j) = \arg \sup_{p_i \in [0,10]} \pi_i(p_i, p_j)$$

2. Make a formal argument that symmetric prices cannot be a pure strategy Nash equilibrium.

3. Make a formal argument that asymmetric prices cannot be a pure strategy Nash equilibrium.

4. Use your result from 1 to find the bound of best response prices.

5. Solve for the mixed strategy equilibrium using pricing bounds using the answer to 1.

6. Solve for the Nash Equilibrium Mixed strategies.

7. Calculate the Nash equilibrium expected profits.